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Classification of Paddy Plant leaf diseases using optimized Support Vector Machine

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Abstract

Plant disease detection is important for several reasons, such as forecasting crop loss, tracking and forecasting infections, identifying host resistance, and researching fundamental biological interactions between hosts and pathogens. Yields can be increased by early identification of plant leaf disease. If automated methods exist to assist farmers in identifying rice illnesses based on images of plant leaves, then this may be feasible. Here, therefore Optimized SVM is proposed. Pre-processing, feature extraction, classification, and segmentation are the four steps of the suggested methodology. The input picture is first converted into red, green, and blue formats. As part of the pre-processing stage, the noise in the green band is eliminated by applying a median filter. Next, utilizing the Texture and Color histogram characteristics, significant features of the green band are retrieved. Following feature extraction, an image is classified as Healthy, Bacterial Blight, Tungro, or Leaf Blast by feeding the retrieved features into the Optimized Support Vector Machine (OSVM) classifier. Adaptive Sunflower Optimization (ASFO) method is used to determine SVM parameters best to maximize SVM performance. A level set segmentation technique is then used to further identify the affected area. Accuracy, sensitivity, and specificity are the metrics used to evaluate how effective our work is. For rice disease data sets and rice leaf disease images, the suggested method's optimized SVM achieved a maximum accuracy of 97.01%.

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Keywords: Plant disease, paddy leaf, blast disease, support vector machine, adaptive sunflower optimization.

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1. Introduction

Everybody's life is touched by agriculture, which is a vital industry that plays a significant role in the growth of the country. In the past, people grew crops on their own property to feed their necessities, but as technology advances, the agriculture industry is now better able to meet the needs of the public. [1-3] With the rapid population growth, there is an increasing need for these inventions to suit the demands of every individual. It can be challenging to produce crops that address a variety of variables, such as soil type, temperature, and humidity. [9-11] Poor malignancy, morphological changes, and plant mortality can be attributed to several factors.

In plants, pathogens and environmental conditions are the main sources of disease. The germs that cause sickness are known as pathogens. Crop diseases are considered a natural part of the ecosystem. It is also one of the many components of the ecosystem that help keep hundreds of thousands of animal species in balance. Crop disease assaults are a major problem in agricultural production because they affect crop output. Food production and quality therefore suffer.

[12–14] Several civilizations, including those of China, India, and Southeast Asia, have evidence of early paddy cropping. First, the seeds are sown in prepared beds. Seedlings are transported to a field or paddy field after twenty to fifty days. After that, reservoirs are placed over it and it is submerged in 5 to 10 cm of water. For the rest of it, it is the season of growth. Farmers spend billions of dollars combating illness. Insufficient technical assistance may often lead to insufficient management of diseases, pollution, and further consequences. Plant diseases that ruin crops may cause famine and starvation. Annual losses for large crops might range from 30 to 50 percent. [4-6] Plants may be influenced by many things at once. The combination of chemicals that cause all plant ailments makes them more complex. Observing plant diseases by hand is a very difficult task. The manual technique requires more labour force, longer processing times, and better understanding of plant diseases. Image processing methods are used to identify plant diseases. Pre-processing, separation, feature extraction, image identification, and final classification are some of the steps involved in photo processing disease.

The collection includes pictures of several diseases being transmitted via leaves. Clustering is the process of organizing an image into groups. The most common diseases that strike paddy plants are leaf spot, wilt rot, brown spots, bacterial rot, and leaf blight.[7-8] Experts can often make a manual diagnosis of plant diseases using just their eyes. sometimes leads to errors in identifying the kind of sickness. The production of rice has decreased recently due to the incapacity to adequately cure illnesses of the paddy leaves. To tackle this rice leaf disease, an appropriate and rapid detection system is required. The main target of this method was leaf blast disease.

The main objective of this research is to identify the kind of paddy disease leaf, separate it from the plant, and determine the affected leaf section. This concept is implemented using the level set technique and the optimum support vector machine (OSVM). For a precise categorization process, feature extraction is provided. The effectiveness of the proposed technique is compared with several methods.

Dilip Kumar Choubey and Ratnesh Kumar Dubey [15] suggested a For farmers, predicting plant diseases is still a difficult task. Thus, a practical strategy is required. As a result, a productive OSVM-based categorization system for paddy leaf diseases has been suggested. Through the use of ASFOA, the parameters are ideally set to enhance the SVM classifier's performance. The SVM and ASFO algorithms' mathematical formulations have been explored in detail. A range of measures have been used to examine the efficacy and performance of the suggested approach in comparison to an alternative approach. The suggested approach's greatest accuracy for predicting plant diseases was 97.54%.

Ratnesh Kumar Dubey and Dilip Kumar Choubey [16] have provided a thorough explanation of the scientific meaning behind the ANN and ASFO computations. The implementation of the recommended plan has been examined using a variety of metrics, and its sufficiency has been contrasted with a unique approach. The suggested approach achieves the highest level of accuracy, 97.94%.

Researchers Ratnesh Kumar Dubey and Dilip Kumar Choubey recently gave a talk on using automation to identify diseases in plant leaves [17]. To achieve this, a combination of a deep learning algorithm and a feature selection approach has been proposed. The SVM-RFE and ARO algorithms have been combined to provide a novel

method for feature selection. The novel feature selection strategy proposed here overcomes the shortcomings of the previous methods. When tested with several categorization criteria, our technique outperformed the state-of-the-art with an enhanced accuracy of 98.86%. The described approach is able to identify agricultural leaf diseases, according to performance metrics. Other crop diseases might potentially be detected using this technology.

Ratnesh Kumar Dubey and Dilip Kumar Choubey have given a talk on automated disease detection in plant leaves [18]. To achieve this, a combination of a deep learning algorithm and a feature selection approach has been proposed. By merging the SVM-RFE and ARF algorithms, a novel method for feature selection has been presented. The novel feature selection strategy proposed here overcomes the shortcomings of the previous methods. We used a number of classification criteria to determine the technique's effectiveness, and the result is an enhanced accuracy of 97.16 percent.

Contribution: - A median filter is used to eliminate noise from the picture before converting it to red, green, and blue formats. The Optimized Support Vector Machine classifier and level set segmentation approach are used to obtain significant features, classify them as Healthy, Bacterial Blight, Tungro, or Leaf Blast, and then optimize them.

Motivation: - Improve agricultural yields and quality, automate diagnosis, detect illnesses early, and more by creating an efficient SVM-based classification system for paddy plant leaf diseases. Fewer financial setbacks Improvements in food safety

Here is how the remainder of the paper is structured: Section 2 provides an analysis of the literature review as well as the research's purpose and contribution. Section 3 lays out the approach that will be used to classify plant diseases; Section 4 presents the findings of the tests; and Section 5 presents the conclusions.

2. Proposed Method Automatic and reliable detection of blast disease in Paddy plant

It is commonly recognized that plant diseases are causing a decline in both rice yield and quality. Crop diseases frequently affect leaves, which lower the quantity and quality of agricultural products. Early identification of leaf diseases can greatly increase output both in terms of quantity and quality. In order to identify illnesses in crops at an early stage of cultivation, this study suggests a machine learning approach for identifying blast disease in paddy fields. The suggested approach would efficiently identify illnesses, lower crop losses, and efficiently boost rice yield. In this work, an OSVM classifier is utilized to detect diseases. In figure 1, the general framework of the suggested method is presented.

The main contribution of this paper is listed below.

- The input photos are noise-reduced using a median filter during pre-processing.
- The next step is to extract textural and color information from every picture.
- The features are sent to the categorization step after feature extraction.
- A trained support vector machine is used for the classification process.
- An adaptive sunflower optimization approach is used to choose the parameters of the suggested support vector machine classifier.
- The impacted area is then segmented using a level-set algorithm after the categorization procedure.
- Multiple indicators are used to assess the efficacy of the suggested approach.

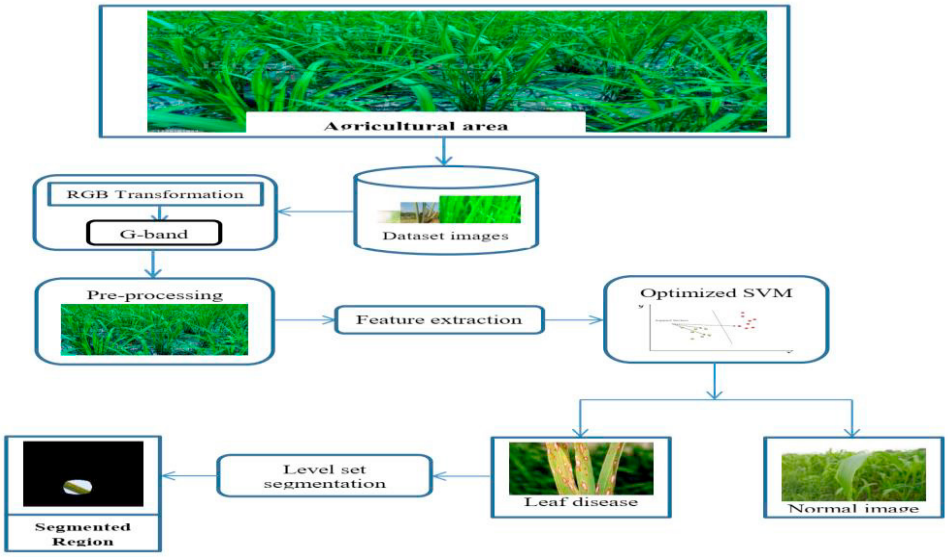

Figure 1: The suggested methodology's overall structure

ASFO based SVM parameter selection algorithm: -

1. Initialization process

    Initialize the SFO parameters

    Initialize the values of C, σ

    Randomly initialize the N_{pop} plants

2. Evaluate the fitness

    Find the best position from the initial population

3. Orientation towards the sun

    Orient all sunflowers towards the sun

4. While $Iter \leq Max_iter$ do

    Compute the step vector for every plant

5. Adjust the position of every plant

6. Remove m(%) plants which further away from the sun

    Find the best solution (C, σ)

End while

2.1 Segmentation using level set algorithm

Pictures with abnormalities (diseases) are sent to the segmentation stage after the classification process. The affected parts of the image are separated at this point. The level set method for segmentation is proposed in this paper. Level set is a useful segmentation technique. This is the pattern and interface identification approach using numbers. The fundamental notion behind the level set technique is to use a partially unique equation (PDE) to generate a level set capacity by treating the definitions as a collection of zero-level abnormal capabilities that are handled in a higher

aspect. Using dynamic bending, distinct components of pictures were introduced via the application of dynamic definitions. For the computation of mathematically practicable edge models, the Euler-Lagrange equation is often used.

The following is a list of the three primary segmentation steps:

1. Initialize $\phi_0 : \Omega \rightarrow \{-1.1\}$

2. Advance: for each grid points: set $\phi^{n+1}(x) = -\phi^n(x)$

If $R\left(\phi^{n+1}, \phi^n, x\right) = 1$;

3. Repeat until $\phi^{n+1} = \phi^n$

$R\left(\phi^{n+1}, \phi^n\right)$ can be interpreted as the logical evaluation of the following inequality:

$$\varepsilon\left(\phi^{n+1}, c_1, c_2; u_0\right) \leq \varepsilon\left(\phi^n, c_1, c_2; u_0\right)$$

3. Results and discussion

The usefulness of the suggested enhanced SVM-based paddy leaf disease classification technique is covered in this section. MATLAB 20a is used to accomplish the suggested methodology. An Intel® Pentium 1.9 GHz CPU, Microsoft® Windows 10 64-bit operating system, and 4 GB of RAM are included in the system configuration. The experimental study uses paddy leaves that were obtained online. Figure 2 shows several sample images.

Figure 2: Experimental used sample images

3.1 Experimental results

The experimental results of proposed paddy leaf disease classification are listed below; Figure 3 depicts the screen short for diseased image output.

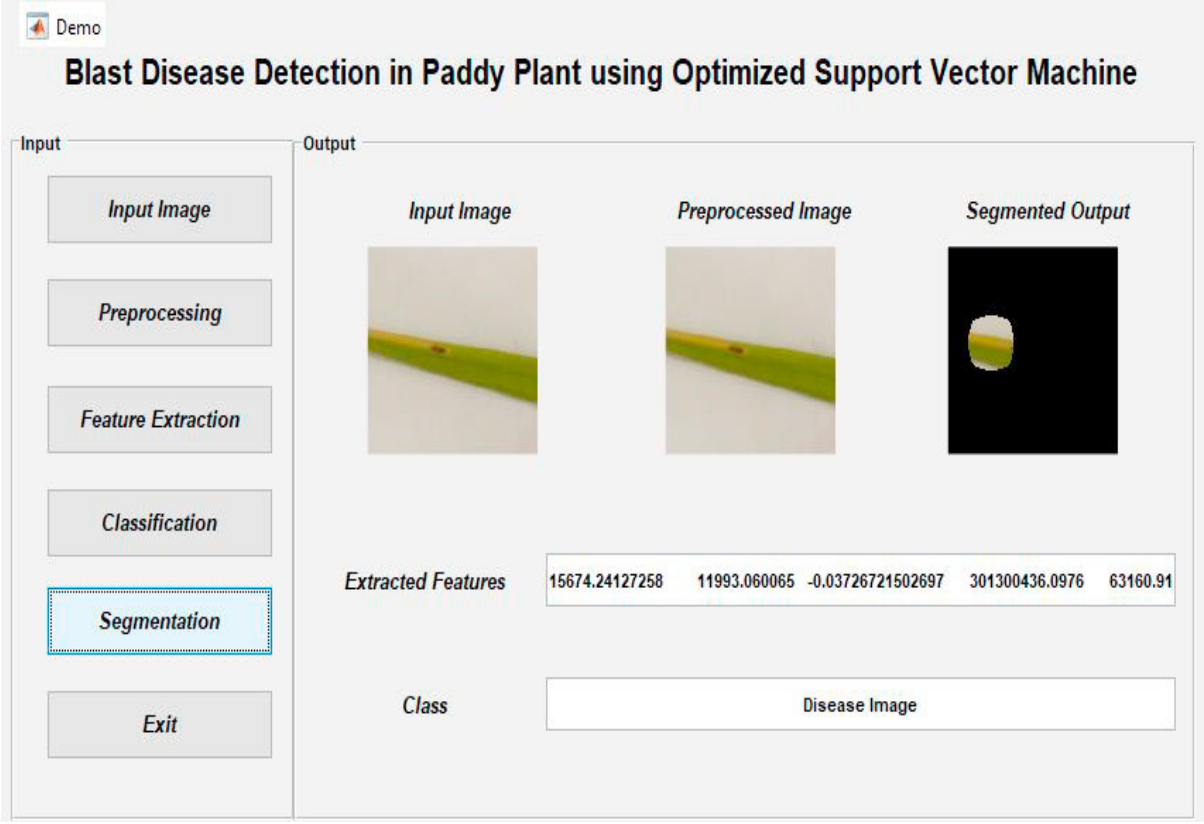

Figure 3: screen short for diseased image output

Figure 4 depicts the screen short for diseased image output.

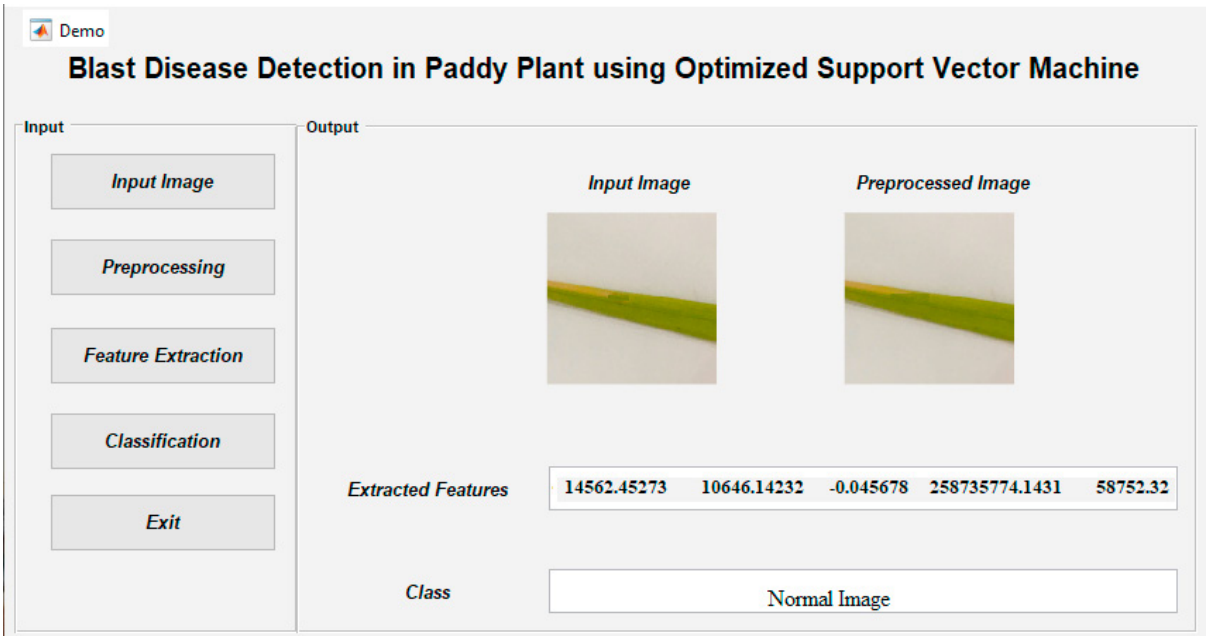

Figure 4: screen short for diseased image output

Figure 5 depicts the Confusion matrix of ASFO+SVM, SVM, KNN and ANN.

Total=314	Normal	Abnormal
Normal	(TP) 151	(FP) 8
Abnormal	(FN) 6	(TN) 149

(a)

Total=314	Normal	Abnormal
Normal	(TP) 150	(FP) 12
Abnormal	(FN) 7	(TN) 145

(b)

Total=314	Normal	Abnormal
Normal	(TP) 133	(FP) 41
Abnormal	(FN) 24	(TN) 116

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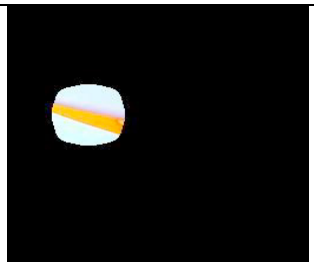
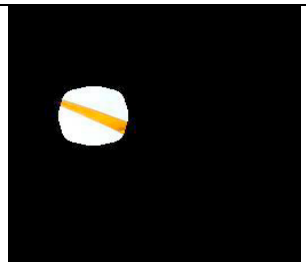
Total=314	Normal	Abnormal
Normal	(TP) 132	(FP) 29
Abnormal	(FN) 25	(TN) 128

(d)

Figure 5: Confusion matrix (a) ASFO+SVM, (b) SVM, (c) KNN and (d) ANN

Figure 6 Depicts the Segmentation stage output.

Input image	Segmented image
	
	

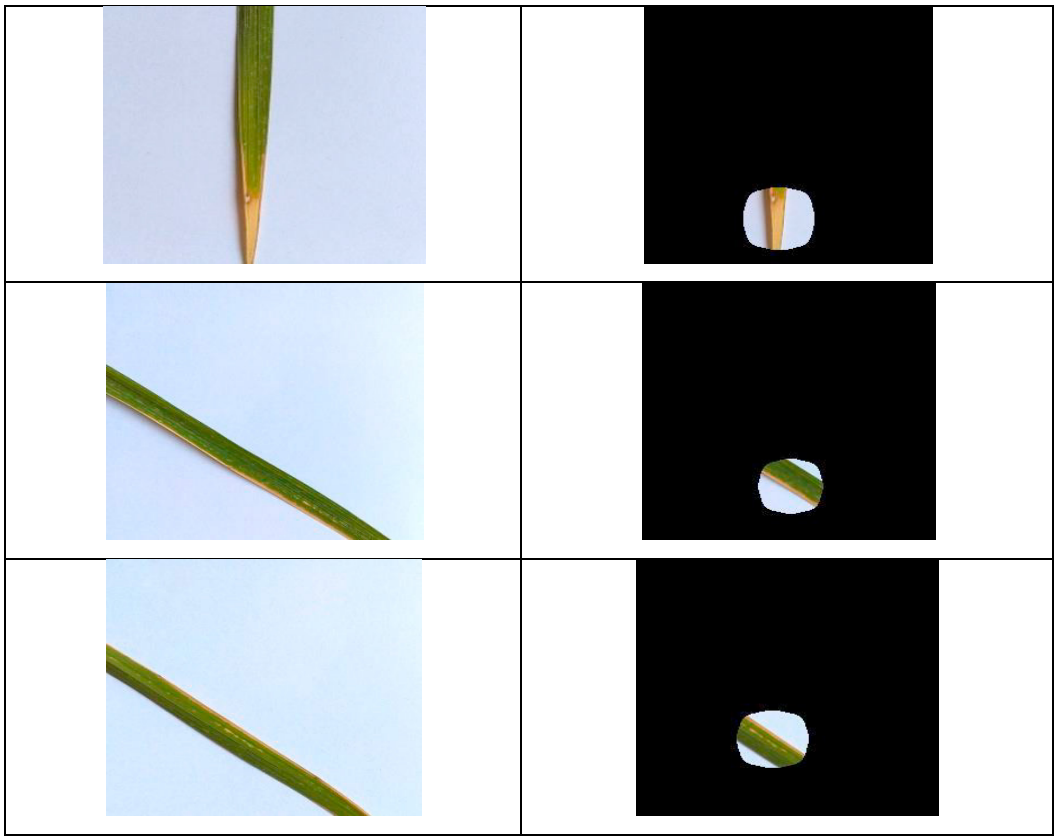


Figure 6: Segmentation stage output

3.2 Comparative analysis output

Several competing methods are compared in order to prove that the suggested way is superior. For this comparison, we used three different methods for illness detection: SVM padding, K-nearest neighbour (KNN), and artificial neural network (ANN). A comparison of efficiency with sensitivity, specificity, and accuracy is shown in Figure 7.

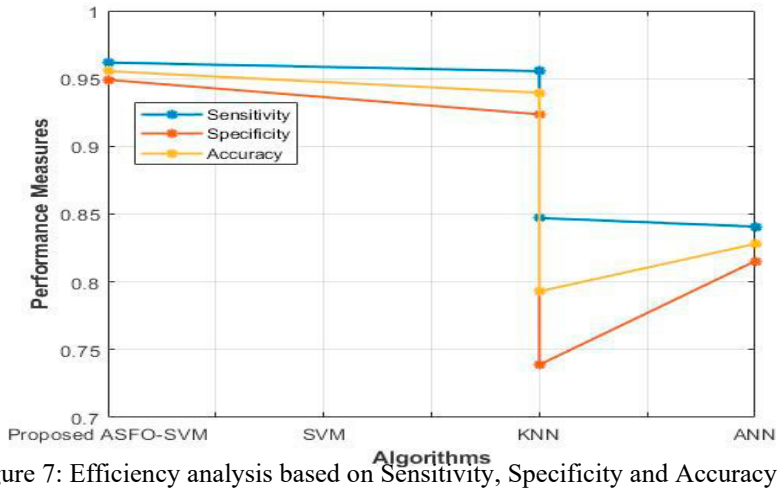


Figure 7: Efficiency analysis based on Sensitivity, Specificity and Accuracy

Figure 8 depicts the Efficiency analysis based on Precision, Recall and F-measure.

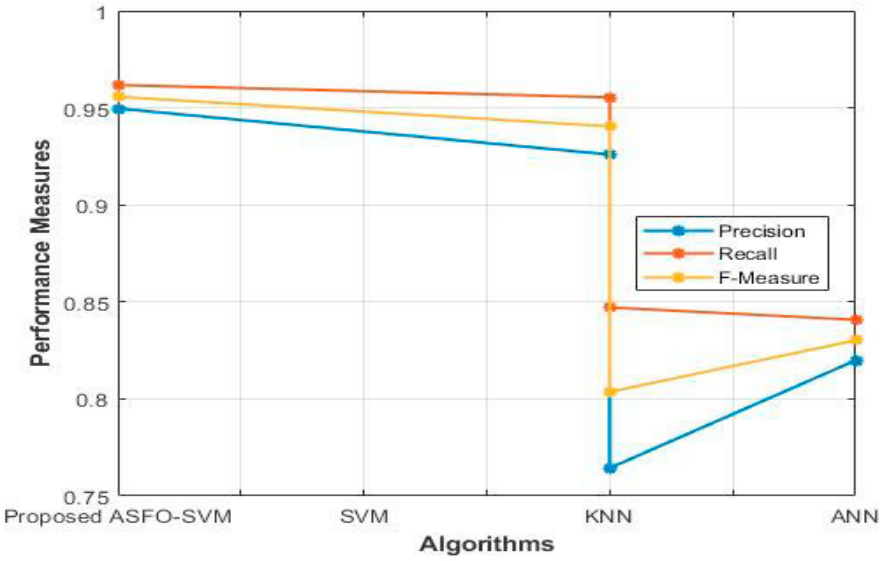


Figure 8: Efficiency analysis based on Precision, Recall and F-measure

Figure 7 uses several classifier efficiency metrics to show how efficient the suggested approach is. Our proposed method yielded a higher accuracy of 97.01% compared to Figure 7, which is 93.94% for SVM-based illness classification, 79.29% for KNN-based disease classification, and 82.8% for ANN-based classification. The highest sensitivity of our proposed technique is 96.17%, which is higher than SVM-based classification by 0.06%, KNN-based classification by 11.91%, and ANN-based illness classification by 12.58%. Figure 7 examines the specificity measure in a similar manner. Figure 7 show that the suggested solution outperformed existing approaches in terms of specificity. Figure 8 shows how successful the suggested method is based on F-measure, accuracy, and recall. Our proposed method beat SVM-based classification by 0.02%, KNN-based classification by 24.2%, and ANN-based classification by 15.83%, as Figure 8 shows. In a similar vein, the figure unequivocally demonstrates that our proposed method produced an F-measure of 95.56% and a maximum recall of 96.17%. Figure 8 makes it abundantly evident that the suggested method produced superior outcomes than the other options. The results section makes it abundantly evident that the recommended method worked better than other options.

4. Conclusion

Farmers still find it difficult to anticipate plant diseases. Consequently, a more effective approach is needed. A successful OSVM-based classification system for paddy leaf diseases has been created. ASFOA is optimally used to select the parameters and boost the SVM classifier's performance. A detailed description of the SVM and ASFO algorithms' mathematical formulas has been provided. Several metrics were used to assess the performance of the suggested approach, and its effectiveness was contrasted with that of other approaches. The recommended method's highest accuracy for plant disease prediction was 97.01%. We will focus on applying deep learning methods to the classification of plant diseases in the future.

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